Week6HW

May 30, 2025

1 Week 6 Assignment

[1]:
```
import matplotlib.pyplot as plt
```

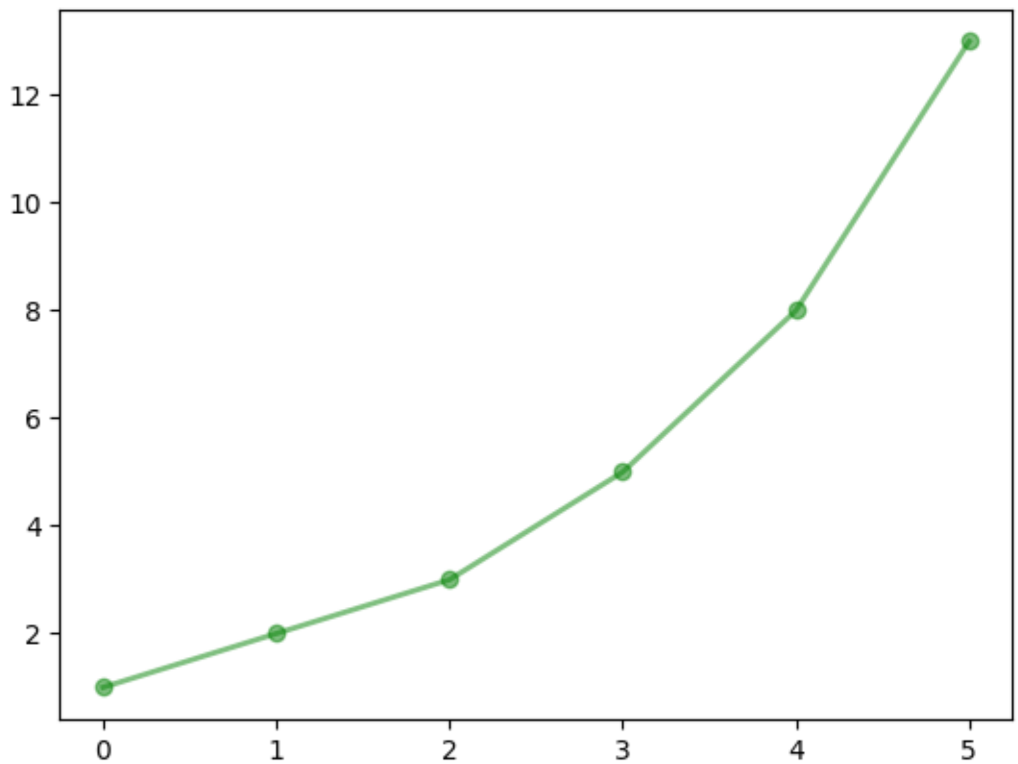
1.1 Problem 1

In the cell below, make a green line plot of the provided list.

[9]:
```
list_to_plot = [1,2,3,5,8,13]
```

[12]:
```
plt.plot(list_to_plot,
         linewidth=2.0,
         linestyle="-",
         color='g',
         alpha=0.5,
         marker='o')
```

[12]:
```
[<matplotlib.lines.Line2D at 0x129875fd0>]
```

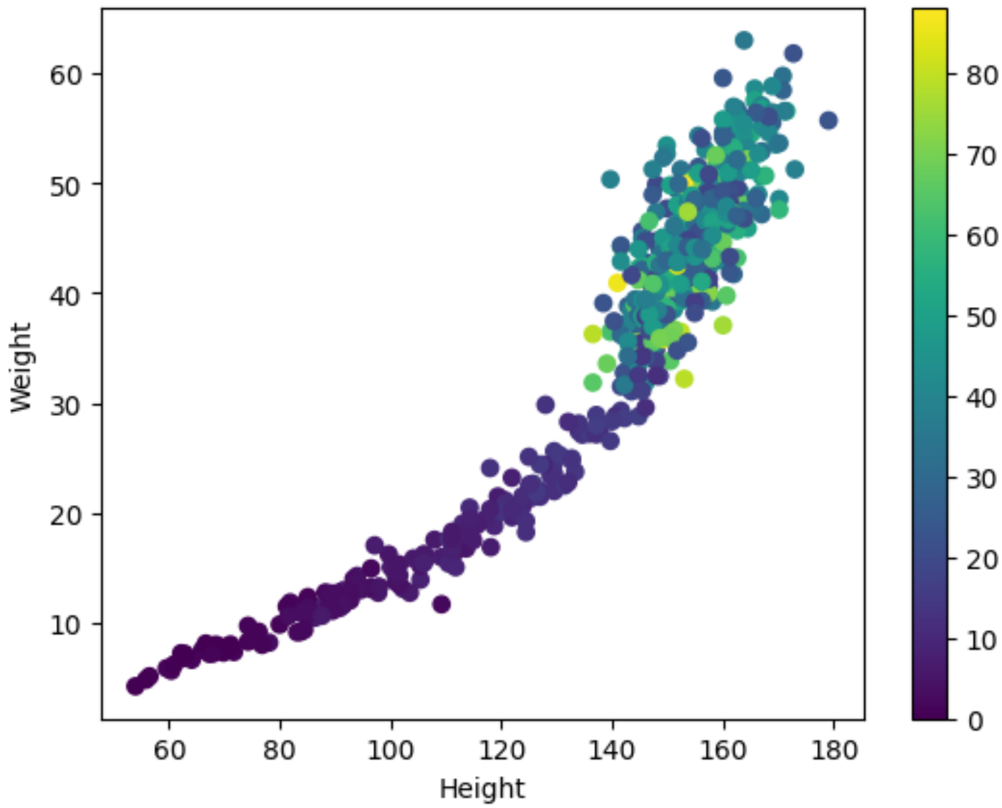


1.2 Problem 2

Review the matplotlib colormaps here. Choose the one you like most and use it to change the look of the following plot.

[46]:
```
import pandas as pd

plt.scatter(HWAS['height'],HWAS['weight'],c=HWAS['age'],cmap = 'viridis')
plt.colorbar()
plt.ylabel('Weight')
plt.xlabel('Height')
plt.show()
```



1.3 Problem 3

In the cell below make a line plot of the closing prices of the GME stock data and line plot of the daily high on the same set of axes. Give the plot a title and label the axes.

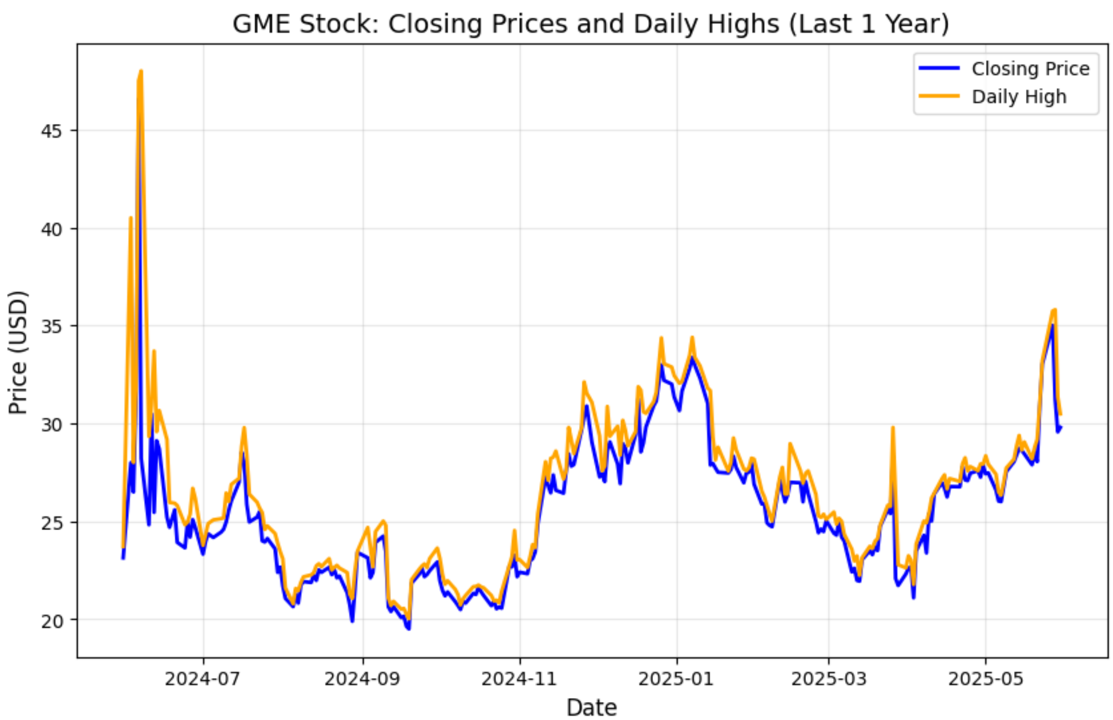
[60]:
```
import yfinance as yf
import matplotlib.pyplot as plt

# Download GME stock data
gme_data = yf.Ticker("GME").history(period="1y")  # Fetch last 1 year of data

# Plot Closing Prices and Daily Highs
plt.figure(figsize=(10, 6))
plt.plot(gme_data.index, gme_data['Close'], label='Closing Price',␣
 ↪color='blue', linewidth=2)
plt.plot(gme_data.index, gme_data['High'], label='Daily High', color='orange',␣
 ↪linewidth=2)

# Add labels, title, legend, and grid
plt.title('GME Stock: Closing Prices and Daily Highs (Last 1 Year)',␣
 ↪fontsize=14)
```

```
plt.xlabel('Date', fontsize=12)
plt.ylabel('Price (USD)', fontsize=12)
plt.legend(fontsize=10)
plt.grid(alpha=0.3)
plt.show()
```



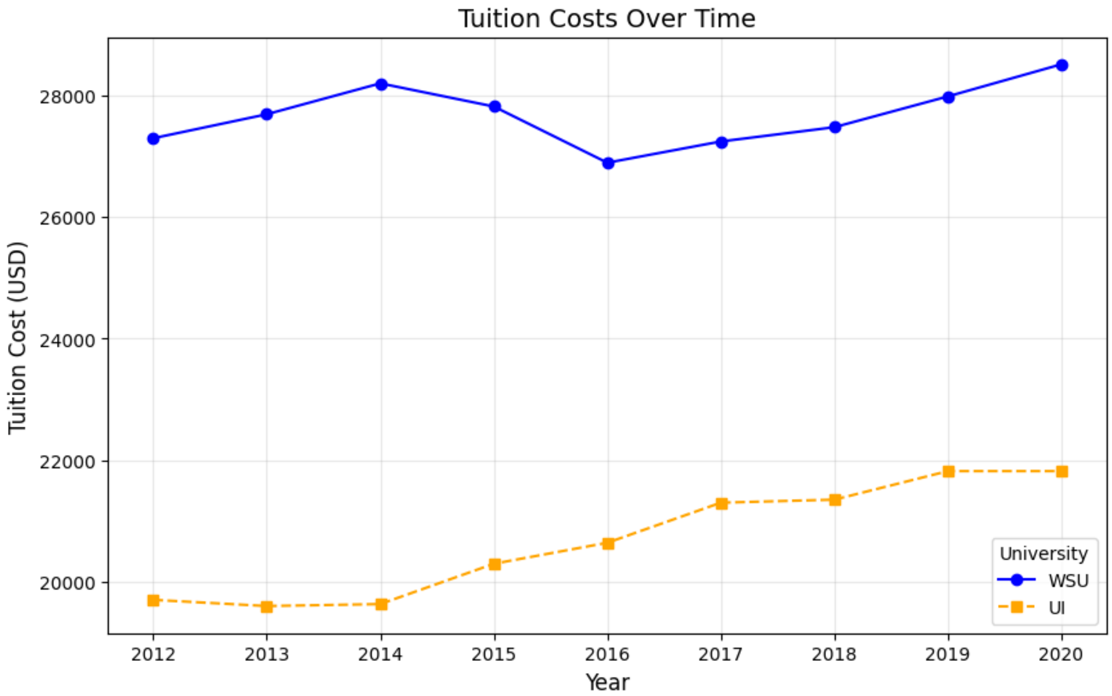
1.4 Problem 4

In the cell below make a scatterplot of the WSU tuition data against the UI tuition data. Color th

[51]:
```
tuition_df =pd.DataFrame({"WSU":
↪[27302,27697,28206,27825,26899,27249,27484,27991,28520],
                          "UI":
↪[19700,19598,19630,20294,20640,21300,21350,21820,21820],
                         "year":[2012,2013,2014,2015,2016,2017,2018,2019,2020]})
```

[58]:
```
# Plot
plt.figure(figsize=(10, 6))
plt.plot(tuition_df['year'], tuition_df['WSU'], label='WSU', marker='o',␣
↪linestyle='-', color='blue')
plt.plot(tuition_df['year'], tuition_df['UI'], label='UI', marker='s',␣
↪linestyle='--', color='orange')
```

```
# Labels and Title
plt.title('Tuition Costs Over Time', fontsize=14)
plt.xlabel('Year', fontsize=12)
plt.ylabel('Tuition Cost (USD)', fontsize=12)
plt.legend(title='University', fontsize=10)
plt.grid(alpha=0.3)
plt.show()
```



1.5 Problem 5

The cells below contain code that generates errors when run. Write a brief description of the problem and modify the code so that the cells exectute properly.

[84]:
```
# It is suppose to be my_dict instead of dict, second was mispelled, amd
# instead of [], it is suppose to be {} given that this is an array
# finally in order print something out we must include print(...)
my_dict = {'first': 1, 'second': 2}
print(my_dict['first'])
```

1

[85]:
```
#Similarly with the previous we must use curly brakets instead of []
a = {1,2,3}
#In order to print the desired value we must set a identifier used to call
#in this cause we used "a_list"
```

```
a_list = list(a)
# Finally it is important to add print(), when we want to print a statement
# In this case we switched 3 to 1 becuase 3 is out of range since we start
# counting from 0 instead of 1, making our max 2 (0,1,2)
print(a_list[2])
```

3

[ ]:
```
# this line is correct since we are not working with an array
my_string = '1235'

#For this line we had instanziate the word result to the string in this case
#the string has 3, which holds the value of 5, resulting in 5^2 = 25
result = int(my_string[3]) ** 2
#The final line will the print out the information stored in result.
print(result)
```

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